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Section I

5 marks

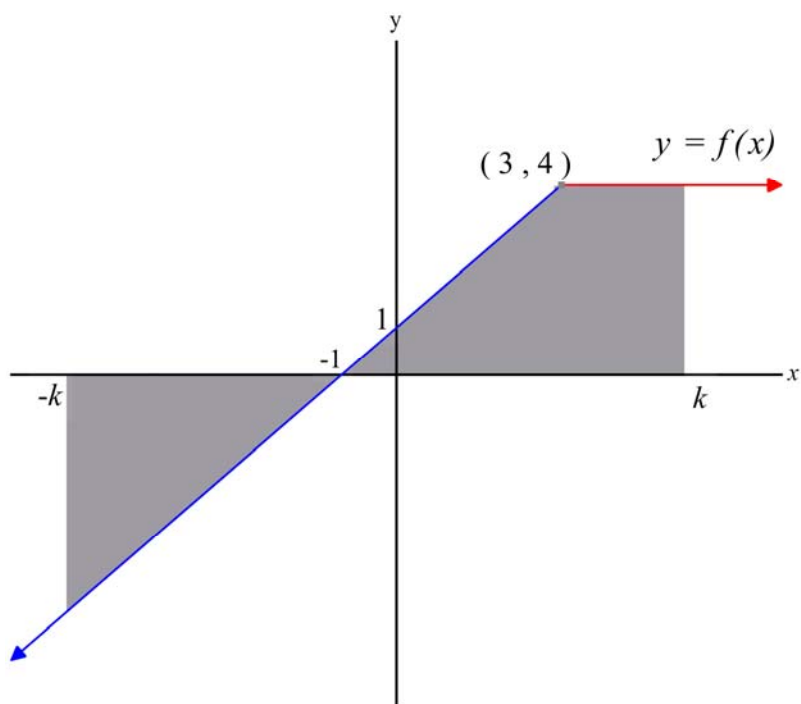
Attempt Questions 1 – 5

Allow about 7 minutes.

Use the multiple choice answer sheet for questions 1-5

1. When the substitution $u = x - 1$ is used, $\int x\sqrt{x-1} \, dx$ is equivalent to:
- (A) $\int (u\sqrt{u} + \sqrt{u}) \, du$
- (B) $\int (u\sqrt{u} + 1) \, du$
- (C) $\int (u\sqrt{u} - \sqrt{u}) \, du$
- (D) $\int (u\sqrt{u} + u) \, du$
2. Which of the following is the formula for the n th term of a series with the sum $S_n = 3n + 2n^2$?
- (A) $T_n = 6 + 4n$
- (B) $T_n = -4 + 9n$
- (C) $T_n = 5 + 4n$
- (D) $T_n = 1 + 4n$
3. What is the value of $\sum_{n=1}^{\infty} \frac{3^{n-1}}{5^{2n-1}}$?
- (A) $\frac{5}{22}$
- (B) $\frac{15}{22}$
- (C) $\frac{25}{122}$
- (D) $\frac{75}{122}$

4. Below is the graph of $y = f(x)$.



What is the value of k so that $\int_{-k}^k f(x) dx = 0$?

- (A) 5
 (B) 8
 (C) 9
 (D) 10
5. A series is defined as $1 + \frac{p^2}{(1-p)^2} + \frac{p^4}{(1-p)^4} + \frac{p^6}{(1-p)^6} + \dots$ where $p \neq 1$ and $p \neq 0$.

For what values p does the series **NOT** have a limiting sum?

- (A) $p \leq -\frac{1}{2}$
 (B) $p \geq \frac{1}{2}$
 (C) $p \leq -\frac{1}{2}$ and $p \geq \frac{1}{2}$
 (D) $-\frac{1}{2} \leq p \leq \frac{1}{2}$

End of Section I

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Section II

28 marks

Attempt Questions 6 - 8

Allow about 43 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In Questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 Marks)

Use a Separate writing booklet

(a) Show that sum of the series $1 + 3 + 5 + 7 + \dots + (2n - 1)$ is a perfect square. 2

(b) Evaluate $\int_1^4 x(2\sqrt{x} - 1) dx$. 2

(c) Find $\int \frac{x^3}{(x^4 + 1)^2} dx$ by using the substitution $u = x^4 + 1$. 3

(d) Use mathematical induction to prove that: 3

$$2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + n(n-1) = \frac{n(n^2 - 1)}{3}$$

for all integers $n \geq 2$.

End of Question 6

Question 7 (9 Marks)

Use a Separate writing booklet.

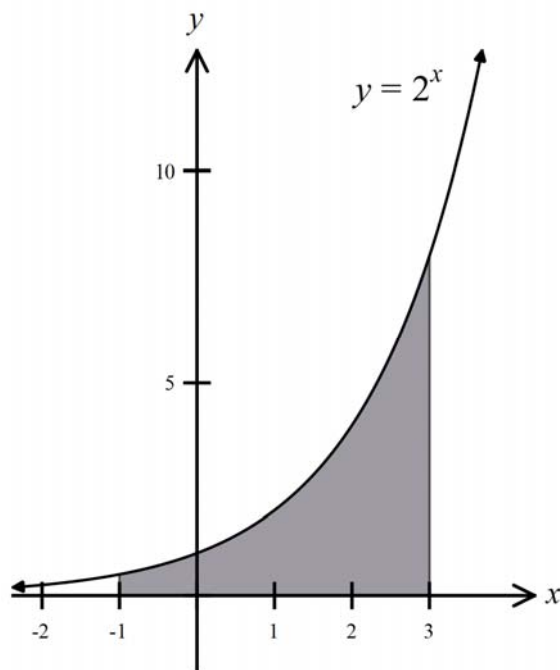
(a) Use the substitution $u^2 = 5 - x$ to evaluate $\int_1^4 (x-1)\sqrt{5-x} \, dx$. **3**

(b) Using the Principle of Mathematical Induction, prove that **3**

$$7^{2n} - 3^{3n} \text{ is divisible by 11}$$

for every integer $n \geq 1$.

(c) In the diagram below, the area under the curve $y = 2^x$ between $x = -1$ and $x = 3$ has been shaded. **3**



Use Simpson's Rule with five function values to determine an approximate volume for the solid of revolution generated when the shaded region is rotated around the x -axis.

Round your answer off to four significant figures.

End of Question 7

Question 8 (9 Marks)

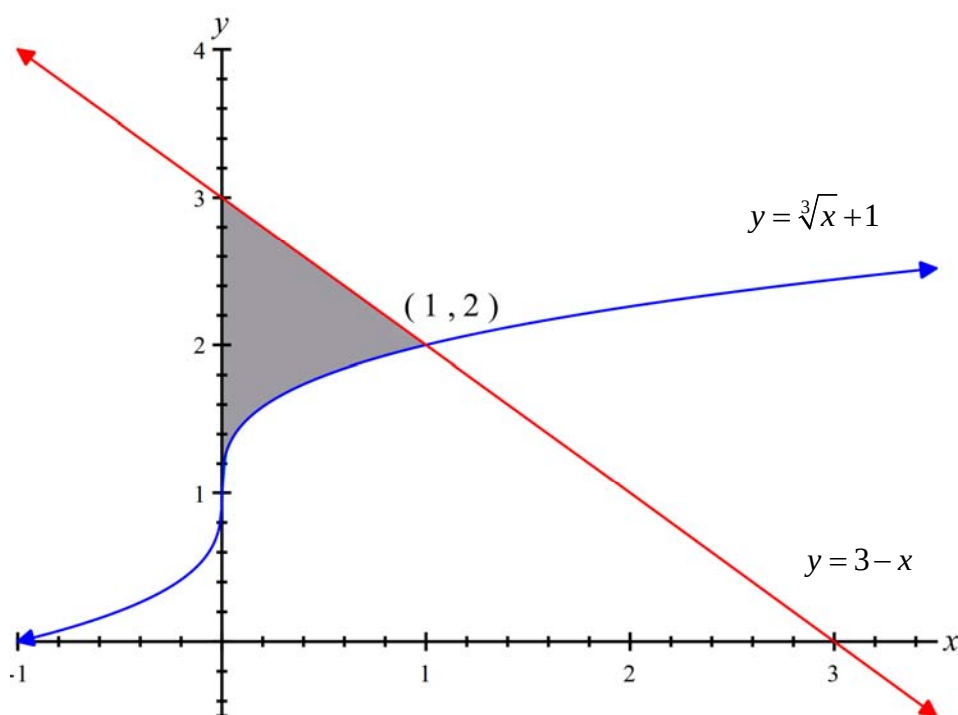
Use a Separate writing booklet

(a) (i) Differentiate $(x-2)(x+4)^7$. Fully factorise your answer. 2

(ii) Hence find $\int (4x-5)(x+4)^6 dx$. 1

(iii) Hence or otherwise find $\int 4x(x+4)^6 dx$. 2

(b) In the diagram below, the line $y = 3 - x$ meets the graph of $y = \sqrt[3]{x} + 1$ at $(1, 2)$.



The region between $y = 3 - x$, $y = \sqrt[3]{x} + 1$ and the y-axis is shaded.
This shaded region is then rotated around the y-axis to form a solid of revolution.

Find the volume of the solid generated. 4

End of Examination

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HSC Examination – Mathematics Extension 1 Task 2 2014

Multiple Choice Answer Sheet

Name _____

Completely fill the response oval representing the most correct answer.

1. A ○ B ○ C ○ D ○
2. A ○ B ○ C ○ D ○
3. A ○ B ○ C ○ D ○
4. A ○ B ○ C ○ D ○
5. A ○ B ○ C ○ D ○

SECTION I – Multiple Choice

1. $\int x\sqrt{x-1} dx$ is equivalent to:

$$u = x - 1$$

$$x = u + 1$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$\int (u+1)\sqrt{u} du = \int (u\sqrt{u} + \sqrt{u}) du$$

Answer is A

2. If $S_n = 3n + 2n^2$, what is T_n

METHOD 1

$$T_n = S_n - S_{n-1}$$

$$= (3(n) + 2(n)^2) - (3(n-1) + 2(n-1)^2)$$

$$= 3n + 2n^2 - 3n + 3 - 2n^2 + 4n - 2$$

$$= 1 + 4n$$

METHOD 2

$$S_1 = 3(1) + 2(1)^2$$

$$= 5$$

$$T_1 = 5$$

and

$$S_2 = 3(2) + 2(2)^2$$

$$= 14$$

$$T_2 = 14 - 5$$

$$= 9$$

$$\therefore a = 5 \text{ and } d = 4$$

$$T_n = 5 + (n-1)4$$

$$= 1 + 4n$$

Answer is D

3. What is the value of $\sum_{n=1}^{\infty} \frac{3^{n-1}}{5^{2n-1}}$?

When $n = 1$

$$\frac{3^0}{5^1} = \frac{1}{5}$$

When $n = 2$

$$\frac{3^1}{5^3} = \frac{3}{125}$$

When $n = 3$

$$\frac{3^2}{5^5}$$

$\therefore \sum_{n=1}^{\infty} \frac{3^{n-1}}{5^{2n-1}}$ is a sum of a geometric series

with first term $\frac{1}{5}$ and common ratio $r = \frac{3}{5^2}$

$$\therefore S_{\infty} = \frac{a}{1-r} = \frac{\frac{1}{5}}{1-\frac{3}{5^2}} = \frac{\frac{1}{5}}{\frac{22}{25}} = \frac{5}{22}$$

Answer is A

4. Find k if $\int_{-k}^k f(x) dx = 0$?

$$AREA_{\text{TRAPEZIUM}} = AREA_{\text{TRIANGLE}}$$

$$\frac{(k+1)+(k-3)}{2} \times 4 = \frac{(k-1)^2}{2}$$

METHOD 1

Test the possible values for k

METHOD 2

Solve the equation for k

$$(4k-4) = \frac{k^2 - 2k + 1}{2}$$

$$8k - 8 = k^2 - 2k + 1$$

$$0 = k^2 - 10k + 9$$

$$0 = (k-9)(k-1)$$

$$k = 9 \text{ or } 1$$

But $k > 3$

$$k = 9$$

Answer is C

5. Not a limiting sum if common ratio r is
 $r \leq -1$ or $r \geq 1$

$$r = \frac{p^2}{(1-p)^2}$$

and $\frac{p^2}{(1-p)^2}$ is always positive or zero.

So solve

$$\frac{p^2}{(1-p)^2} \geq 1$$

$$p^2 \geq (1-p)^2$$

$$p^2 \geq 1 - 2p + p^2$$

$$0 \geq 1 - 2p$$

$$2p \geq 1$$

$$p \geq \frac{1}{2}$$

Answer is B

QUESTION 6

(a)

$$1+3+5+7+\dots+(2n-1)$$

$$a=1 \quad d=2$$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_n = \frac{n}{2}[2(1) + (n-1)2]$$

$$S_n = \frac{n}{2}[2n]$$

$$S_n = n^2$$

- (b) Evaluate $\int_1^4 x(2\sqrt{x}-1) dx$.

$$\int_1^4 x(2\sqrt{x}-1) dx$$

$$\int_1^4 (2x\sqrt{x}-x) dx$$

$$\int_1^4 \left(2x^{\frac{3}{2}}-x\right) dx$$

$$\left[\frac{4x^{\frac{5}{2}}}{5}-\frac{x^2}{2}\right]_1^4$$

$$\left(\frac{4(4)^{\frac{5}{2}}}{5}-\frac{(4)^2}{2}\right)-\left(\frac{4(1)^{\frac{5}{2}}}{5}-\frac{(1)^2}{2}\right)$$

$$\left(\frac{128}{5}-8\right)-\frac{8-5}{10}$$

$$\frac{173}{10}$$

- (c) Find $\int \frac{x^3}{(x^4+1)^2} dx$ by using $u = x^4 + 1$.

$$\int \frac{x^3}{(x^4+1)^2} dx$$

$$u = x^4 + 1$$

$$\frac{du}{dx} = 4x^3$$

$$\int \frac{1}{4(u)^2} du$$

$$dx = \frac{du}{4x^3}$$

$$\int \frac{1}{4(u)^2} du = \int \frac{u^{-2}}{4}$$

$$= -\frac{u^{-1}}{4} + C$$

$$= -\frac{1}{4u} + C$$

$$= -\frac{1}{4(x^4+1)} + C$$

(d)

$$LHS = 2 \times 1 = 2$$

$$RHS = \frac{2(2^2-1)}{3} = 2$$

$\therefore$ True for $n = 2$

Assume true for $n = k$

$$2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) = \frac{k(k^2-1)}{3}$$

Prove that if true for $n = k$

then true for $n = k + 1$

AIM : prove that:

$$\begin{aligned}
 & 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) + (k+1)k \\
 &= \frac{(k+1)((k+1)^2 - 1)}{3} \\
 &= \frac{(k+1)(k^2 + 2k)}{3} \\
 &= \frac{k(k+1)(k+2)}{3} \\
 LHS &= 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) + (k+1)k \\
 &= \frac{k(k^2 - 1)}{3} + (k+1)k \\
 &= \frac{k(k-1)(k+1)}{3} + (k+1)k \\
 &= \frac{(k+1)k}{3} [k-1+3] \\
 &= \frac{(k+1)k}{3} [k+2] = RHS
 \end{aligned}$$

Question 7 (9 Marks)

(a) Use $u^2 = 5 - x$ to evaluate $\int_1^4 (x-1)\sqrt{5-x} \, dx$

$$\begin{aligned}
 u^2 &= 5 - x \\
 x &= 5 - u^2 \\
 \frac{dx}{du} &= -2u \\
 dx &= (-2u) du \\
 x=4 \quad u &= \sqrt{5-4} = 1 \\
 x=1 \quad u &= \sqrt{5-1} = 2 \\
 \int_1^4 (x-1)\sqrt{5-x} \, dx &= \int_2^1 (5-u^2-1)\sqrt{u^2} (-2u) du \\
 &= -2 \int_2^1 (4-u^2)(u)(u) du \\
 &= -2 \int_2^1 (4u^2 - u^4) du \\
 &= -2 \left[\frac{4u^3}{3} - \frac{u^5}{5} \right]_2^1 \\
 &= -2 \left[\left(\frac{4(1)^3}{3} - \frac{(1)^5}{5} \right) - \left(\frac{4(2)^3}{3} - \frac{(2)^5}{5} \right) \right] \\
 &= 6 \frac{4}{15}
 \end{aligned}$$

(b) Show that the result is true for $n = 2$

$$\begin{aligned}
 7^{2(1)} - 3^{3(1)} &= 49 - 27 \\
 &= 22
 \end{aligned}$$

Which is divisible by 11

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$7^{2k} - 3^{3k} = 11M \quad \text{Where } M \text{ is an integer.}$$

$$7^{2k} = 11M + 3^{3k}$$

Prove that if true for $n = k$

then true for $n = k + 1$

AIM : prove that:

$$7^{2(k+1)} - 3^{3(k+1)} = 11N \quad \text{Where } N \text{ is an integer.}$$

$$\begin{aligned}
 LHS &= 7^{2(k+1)} - 3^{3(k+1)} \\
 &= 7^{2k+2} - 3^{3k+3} \\
 &= 49 \times 7^{2k} - 27 \times 3^{3k} \\
 &= 49 \times (11M + 3^{3k}) - 27 \times 3^{3k} \quad \text{From assumption} \\
 &= 49 \times 11M + 49 \times 3^{3k} - 27 \times 3^{3k} \\
 &= 49 \times 11M + 22 \times 3^{3k} \\
 &= 11(49M + 2 \times 3^{3k}) \\
 &= 11N \quad \because (49M + 2 \times 3^{3k}) \text{ is an integer}
 \end{aligned}$$

(c) Use Simpson's Rule with five function values to determine an approximate volume.

x	-1	0	1	2	3
2^x	$\frac{1}{2}$	1	2	4	8
2^{2x}	$\frac{1}{4}$	1	4	16	64

k	1	4	2	4	1
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$k2^{2x}$	$\frac{1}{4}$	4	8	64	64
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$$\sum ky^2 = 140\frac{1}{4}$$

$$\begin{aligned} \text{Volume} &= \pi \int_{-1}^3 y^2 dx \\ &= \pi \frac{1}{3} \left(140 \frac{1}{4} \right) \\ &= \pi \times 46 \frac{3}{4} \\ &= \frac{187\pi}{4} = 146.9 \text{ units}^3 \end{aligned}$$

Question 8

$$\begin{aligned} \text{(a)} \quad \frac{d}{dx} (x-2)(x+4)^7 &= (x+4)^7 + 7(x-2)(x+4)^6 \\ &= (x+4)^6 [x+4 + 7x-14] \\ &= (x+4)^6 (8x-10) \end{aligned}$$

$$\text{(b) (ii) Hence find } \int (4x-5)(x+4)^6 dx.$$

$$\int (x+4)^6 (8x-10) = (x-2)(x+4)^7 \quad \text{From part (a)}$$

$$\int (x+4)^6 (4x-5) = \frac{1}{2} (x-2)(x+4)^7$$

$$\text{(iii) Hence find } \int 4x(x+4)^6 dx.$$

$$\int (4x-5)(x+4)^6 = \frac{1}{2} (x-2)(x+4)^7 + C \quad \text{From part (b)}$$

$$\int 4x(x+4)^6 - \int 5(x+4)^6 = \frac{1}{2} (x-2)(x+4)^7 + C$$

$$\begin{aligned} \int 4x(x+4)^6 &= \frac{1}{2} (x-2)(x+4)^7 + \int 5(x+4)^6 + C \\ &= \frac{1}{2} (x-2)(x+4)^7 + \frac{5(x+4)^7}{7} + D \end{aligned}$$

(b) Find the volume of the solid generated.

$$\begin{aligned} y &= 3-x & y &= \sqrt[3]{x}+1 \\ x &= 3-y & y-1 &= \sqrt[3]{x} \\ & & (y-1)^3 &= x \end{aligned}$$

$$\begin{aligned}
V &= \pi \int_2^3 (3-y)^2 dy + \pi \int_1^2 (y-1)^6 dy \\
&= \pi \left[-\frac{(3-y)^3}{3} \right]_2^3 + \pi \left[\frac{(y-1)^7}{7} \right]_1^2 \\
&= \pi \left[\left(-\frac{(3-(3))^3}{3} \right) - \left(-\frac{(3-(2))^3}{3} \right) \right] + \pi \left[\left(\frac{((2)-1)^7}{7} \right) - \left(\frac{((1)-1)^7}{7} \right) \right] \\
&= \pi \left[(0) + \left(\frac{1}{3} \right) \right] + \pi \left[\left(\frac{1}{7} \right) - (0) \right] \\
&= \frac{10\pi}{21} \text{Units}^3
\end{aligned}$$